

EXPERIMENT NO:-04

Aim: Apply DML Commands for the specified system.

Theory:

The structured query language (SQL) commands deal with the manipulation of data present in the database that belongs to the DML or Data Manipulation Language. This includes most of the SQL statements.

Examples of DML

The examples of DML in the Database Management System (DBMS) are as follows –

- SELECT – Retrieve data from the database.
- INSERT – Insert data into a table.
- UPDATE – Update existing data within a table.
- DELETE – Delete records from a database table.

Syntax for DML Commands

The syntax for the DML commands are as follows –

- **INSERT**

Insert command is used to insert data into a table.

The syntax for insert command is as follows –

Syntax

Insert into <table_name> (column list) values (column values);

For example, if we want to insert multiple rows to the Employee table, we can use the following command –

Example

Insert into Employee(Emp_id, Emp_name) values (001, “ bhanu”);

Insert into Employee(Emp_id, Emp_name) values (002, “ hari”);

Insert into Employee(Emp_id, Emp_name) values (003, “ bob”);

- **SELECT**

Select command is used to retrieve data from the database.

The syntax for the select command is as follows –

Syntax

SELECT * from <table_name>;

For example, if we want to select all rows from the Employee database, we can use the following command –

Example

SELECT * from Employee;

- **DELETE**

Delete command is used to delete records from a database table.

The syntax for the delete command is as follows -

Syntax

Delete from <table_name>WHERE condition;

For example, if we want to delete an entire row of employee id 002, we can use the following command –

Example

DELETE from Employee WHERE Emp_id=002;

- **UPDATE**

Update command is used to update existing data within a table.

The syntax for the update command is as follows -

Syntax

UPDATE <table_name> SET column_number =value_number WHERE condition;

For example, if we want to update the name of the employee having the employee id 001, we can use the command given below –

Example

UPDATE Employee SET Emp_name= Ram WHERE Emp_id= 001;

Application of DML commands on Specified Case study with Input & Output:

Command :- INSERT : to add a new record in a table.

Sample Insert Example on EMP table:

```
insert into emp values(7844,'TURNER','SALESMAN',7698,STR_TO_DATE('08-09-1981', '%d-%m-%Y'),1500,0,30);
```

```
insert into emp values(7876,'ADAMS','CLERK',7788,STR_TO_DATE('23-05-1981', '%d-%m-%Y'),1100,NULL,20);
```

```
insert into emp values(7900,'JAMES','CLERK',7698,STR_TO_DATE('03-12-1981', '%d-%m-%Y'),950,NULL,30);
```

Sample Insert Example on DEPT table:

```
insert into dept values(10,'ACCOUNTING','NEWYORK');
```

```
insert into dept values(20,'RESEARCH','DALLAS');
```

```
mysql> SELECT * FROM EMP;
```

eno	ename	job	mgr	hiredate	sal	comm	dno
7369	SMITH	CLERK	7902	1980-12-07	800.00	NULL	20
7499	ALLEN	SALESMAN	7698	2020-02-19	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7566	JONES	MANAGER	7839	1981-04-22	2975.00	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1987-04-19	3000.00	NULL	20
7839	KING	PRESIDENT	1	1981-09-08	5000.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7876	ADAMS	CLERK	7788	1981-05-23	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

14 rows in set (0.00 sec)

```
mysql> select * from dept;
```

dno	dname	loc
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

Command :- Update : is to update the a column value of a table.

Q1: Give all employee a commission of Rs. 500 .

Ans : UPDATE emp SET comm = 500;

Q2: Increase everybody's salary by 10%.

Ans : UPDATE emp SET sal= sal + (sal *.1);

Q3: Increase salary of employee no 7654 by 10% .

Ans : UPDATE emp SET sal= sal + (sal *.1) where eno = 7654;

Q4: Increase salary of employee's by 10% who are working as a 'CLERK'.

Ans : UPDATE emp SET sal= sal + (sal *.1) where job = 'CLERK';

Command :- Delete: is use to delete row from table.

Q5. Remove all employees of the company.

Ans : Delete from emp;

Q6. Remove employee's working as 'CLERK' from emp.

Ans : Delete from emp where job = 'CLERK';

Command :-

Select: is use to retrieve information(rows) from table.

Syntax :

SELECT <col-List> FROM <Table-name>

[WHERE <condition>]

[GROUP BY <col-name>]

[HAVING <condition>]

[ORDER BY <expression>;

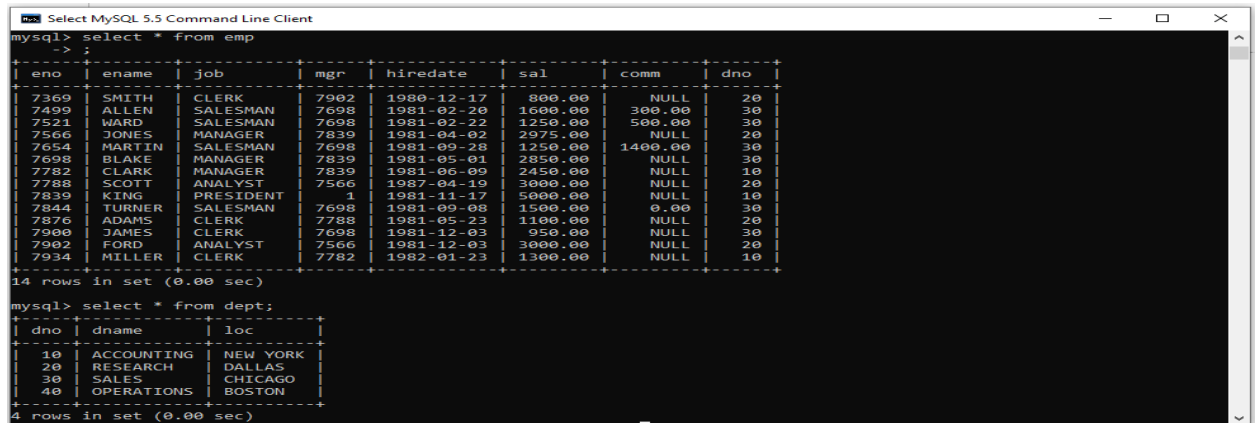
Examples :

Q7: Display all employee's details.

Ans: select * from emp;

Q8 : Display details of all Department.

Ans: select * from dept;



The screenshot shows the MySQL 5.5 Command Line Client interface. The first query, 'select * from emp;', returns 14 rows of employee data. The second query, 'select * from dept;', returns 4 rows of department data.

eno	ename	job	mgr	hiredate	sal	comm	dno
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600.00	300.00	30
7521	WARD	SALESMAN	7698	1981-02-22	1250.00	500.00	30
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250.00	1400.00	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1987-04-19	3000.00	NULL	20
7839	KING	PRESIDENT	1	1981-11-17	5000.00	NULL	10
7844	TURNER	SALESMAN	7698	1981-09-08	1500.00	0.00	30
7876	ADAMS	CLERK	7788	1981-05-23	1100.00	NULL	20
7900	JAMES	CLERK	7698	1981-12-03	950.00	NULL	30
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20
7934	MILLER	CLERK	7782	1982-01-23	1300.00	NULL	10

14 rows in set (0.00 sec)

dno	dname	loc
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

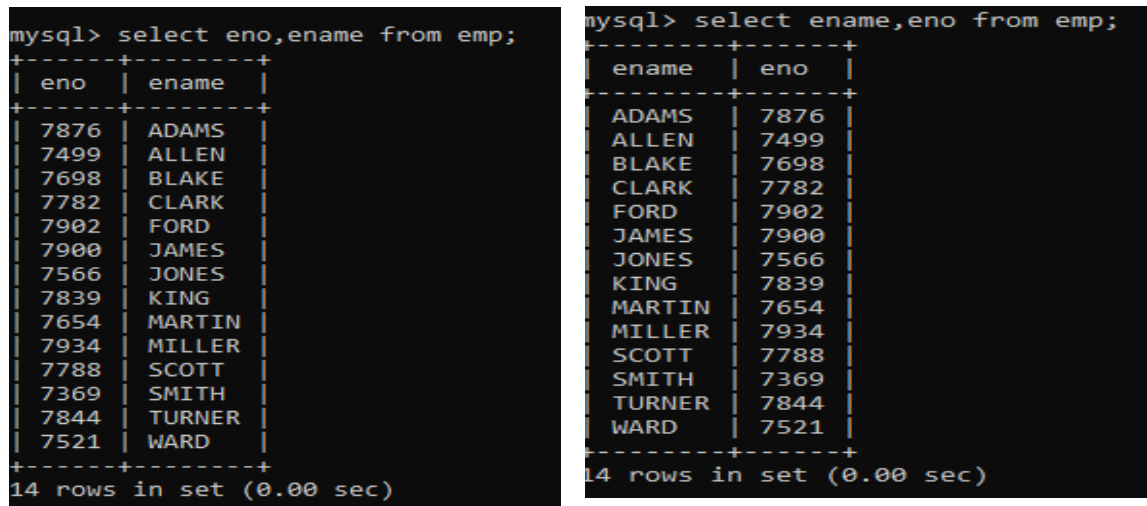
4 rows in set (0.00 sec)

Q9: Display only emp. Number and name of the employee of all employee's .

Ans: select eno, ename from emp;

Q10: Display only emp. Name and emp number of the employee of all employee's .

Ans: select ename, eno from emp;



The left screenshot shows the output of 'select eno, ename from emp;', displaying employee numbers and names. The right screenshot shows the output of 'select ename, eno from emp;', displaying employee names and numbers.

eno	ename
7876	ADAMS
7499	ALLEN
7698	BLAKE
7782	CLARK
7902	FORD
7900	JAMES
7566	JONES
7839	KING
7654	MARTIN
7934	MILLER
7788	SCOTT
7369	SMITH
7844	TURNER
7521	WARD

14 rows in set (0.00 sec)

ename	eno
ADAMS	7876
ALLEN	7499
BLAKE	7698
CLARK	7782
FORD	7902
JAMES	7900
JONES	7566
KING	7839
MARTIN	7654
MILLER	7934
SCOTT	7788
SMITH	7369
TURNER	7844
WARD	7521

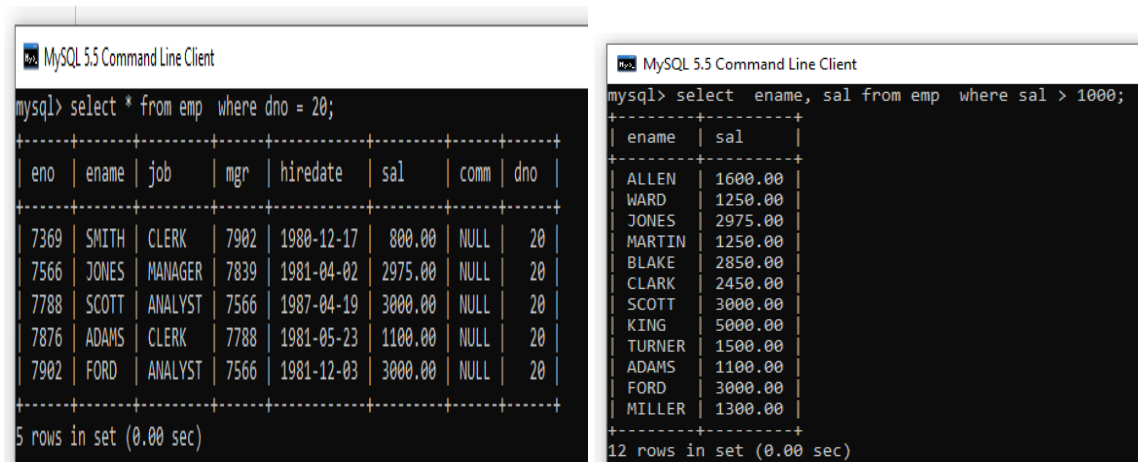
14 rows in set (0.00 sec)

Q11: List the employee details belonging to the department 20;

Ans: select * from emp where dno = 20;

Q12: List the name and salary of the employees whose salary is more than 1000.

Ans: select ename, sal from emp where sal > 1000;



The image shows two side-by-side screenshots of the MySQL 5.5 Command Line Client. The left screenshot shows the command 'mysql> select * from emp where dno = 20;' and its output, which is a table with 8 columns: eno, ename, job, mgr, hiredate, sal, comm, and dno. It contains 5 rows of data. The right screenshot shows the command 'mysql> select ename, sal from emp where sal > 1000;' and its output, which is a table with 2 columns: ename and sal. It contains 12 rows of data.

eno	ename	job	mgr	hiredate	sal	comm	dno
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7788	SCOTT	ANALYST	7566	1987-04-19	3000.00	NULL	20
7876	ADAMS	CLERK	7788	1981-05-23	1100.00	NULL	20
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20

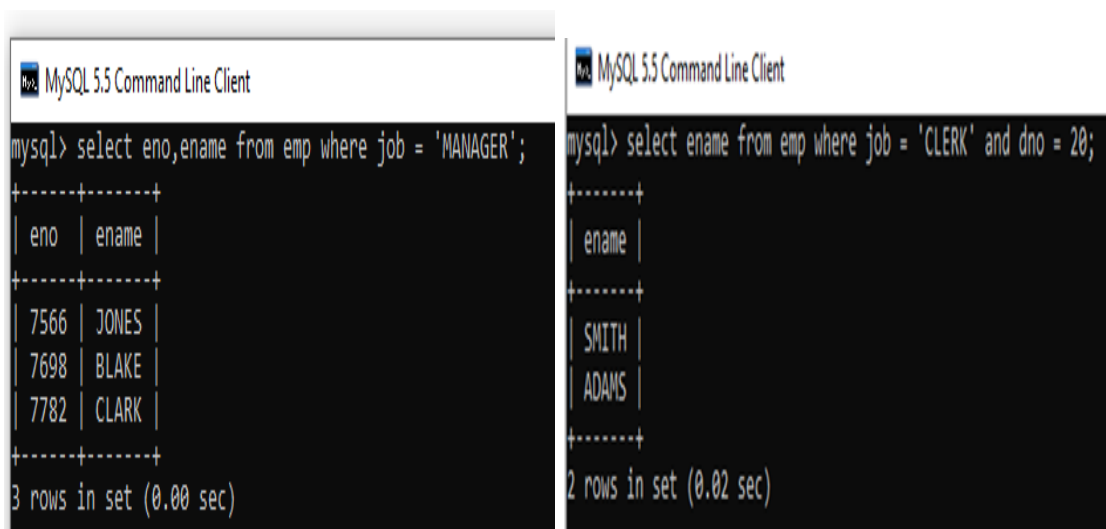
ename	sal
ALLEN	1600.00
WARD	1250.00
JONES	2975.00
MARTIN	1250.00
BLAKE	2850.00
CLARK	2450.00
SCOTT	3000.00
KING	5000.00
TURNER	1500.00
ADAMS	1100.00
FORD	3000.00
MILLER	1300.00

Q13: List the emp. number and name of manager.

Ans: select eno, ename from emp where job = 'MANAGER';

Q14: List the names pf the clerk working in department 20.

Ans: select ename from emp where job = 'CLERK' and dno = 20;



The image shows two side-by-side screenshots of the MySQL 5.5 Command Line Client. The left screenshot shows the command 'mysql> select eno,ename from emp where job = 'MANAGER';' and its output, which is a table with 2 columns: eno and ename. It contains 3 rows of data. The right screenshot shows the command 'mysql> select ename from emp where job = 'CLERK' and dno = 20;' and its output, which is a table with 1 column: ename. It contains 2 rows of data.

eno	ename
7566	JONES
7698	BLAKE
7782	CLARK

ename
SMITH
ADAMS

Q15: List the names of Analyst and salesman.

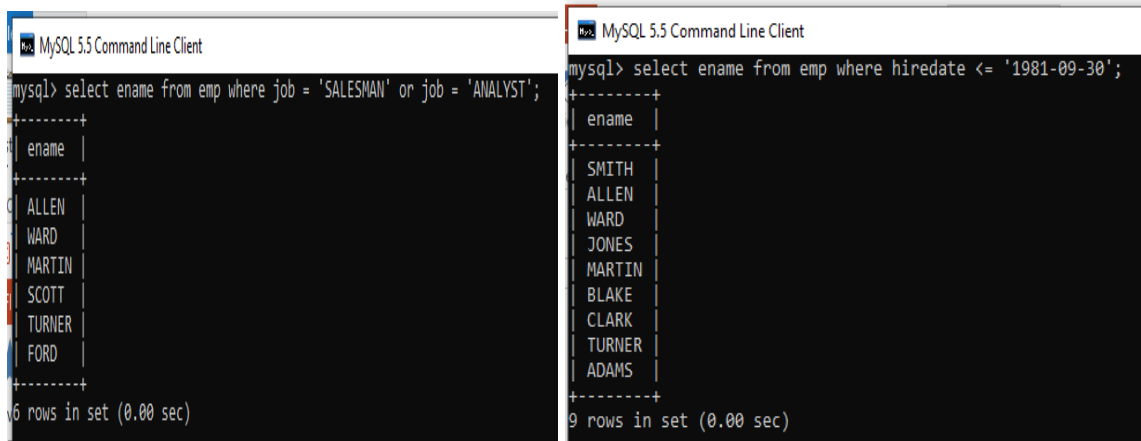
Ans: select ename from emp where job = 'SALESMAN' or job = 'ANALYST';

Q16: List the details of the employee who have joined before the end of Sept. 81.

Ans: select * from emp where hiredate <= '1981-09-30';

Q17 : List the names of employees who are not Managers.

Ans: select ename from emp where job <> 'MANAGER';



The image shows two side-by-side screenshots of the MySQL 5.5 Command Line Client. The left screenshot shows the command 'mysql> select ename from emp where job = 'SALESMAN' or job = 'ANALYST';' and its output, which lists the names of six employees: ALLEN, WARD, MARTIN, SCOTT, TURNER, and FORD. The right screenshot shows the command 'mysql> select ename from emp where hiredate <= '1981-09-30';' and its output, which lists the names of nine employees: SMITH, ALLEN, WARD, JONES, MARTIN, BLAKE, CLARK, TURNER, and ADAMS. Both screenshots show the output in a table format with a header row and a footer row indicating the number of rows in the set.

Use of Special operator:

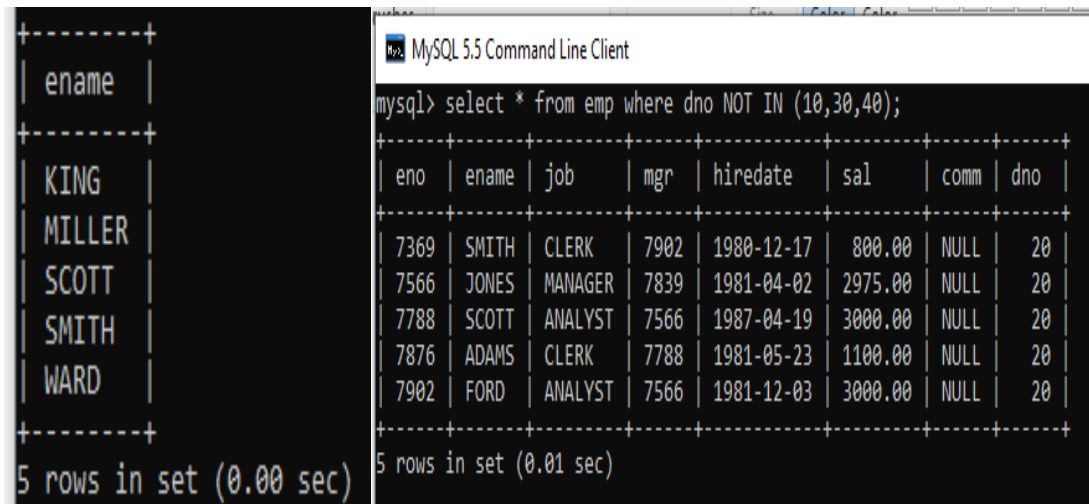
Op. No.	Name of Special operator	Purpose
1	IN	Checking a value is member of Set
2	BETWEEN	Checking a values within a range
3	LIKE	Matching a pattern from a column % - represent a sequence of character _ (underscore) – stands for any single character

Q18: List the names of the employee whose employee numbers are 7369, 7521, 7839, 7934, 7788:

Ans: Select ename from emp where eno IN (7369,7521,7839,7788);

Q19: List the employee details not belonging to the dept. 10, 30 and 40.

Ans : Select * from emp where dno NOT IN (10,30,40);

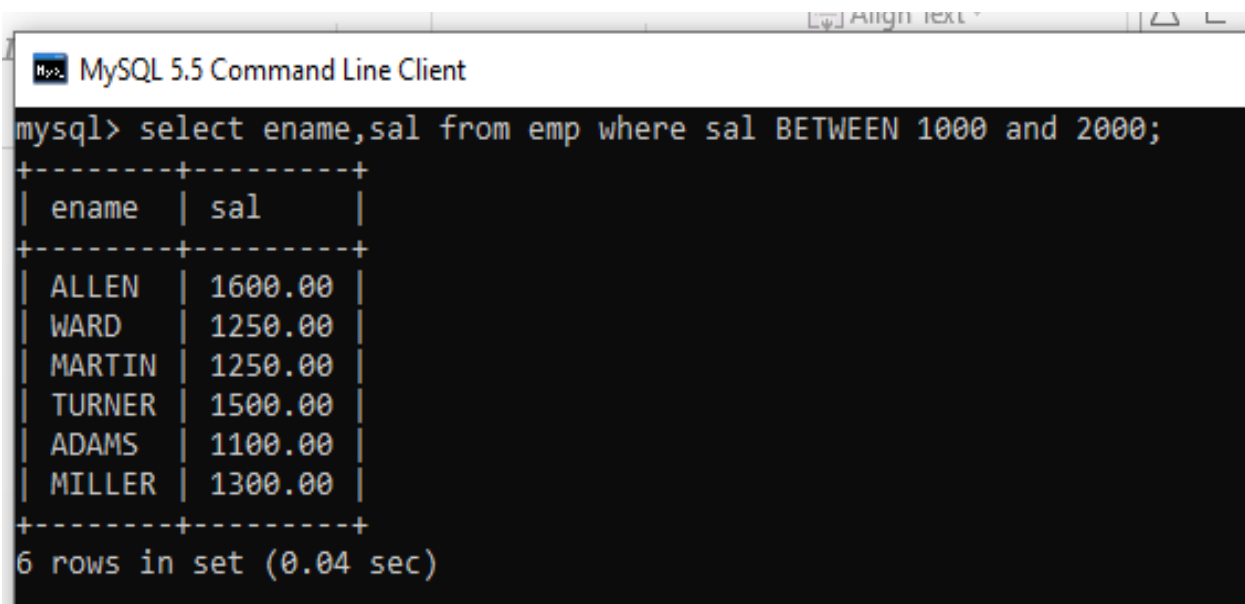


The screenshot shows the MySQL 5.5 Command Line Client interface. The query executed is `mysql> select * from emp where dno NOT IN (10,30,40);`. The result is a table with 9 columns: eno, ename, job, mgr, hiredate, sal, comm, and dno. There are 5 rows of data displayed. Below the table, it says "5 rows in set (0.00 sec)".

eno	ename	job	mgr	hiredate	sal	comm	dno
7369	SMITH	CLERK	7902	1980-12-17	800.00	NULL	20
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7788	SCOTT	ANALYST	7566	1987-04-19	3000.00	NULL	20
7876	ADAMS	CLERK	7788	1981-05-23	1100.00	NULL	20
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20

Q20 : List the employee name and salary whose salary is in the range of 1000 and 2000.

Ans : Select ename, sal from emp where sal BETWEEN 1000 and 2000;



The screenshot shows the MySQL 5.5 Command Line Client interface. The query executed is `mysql> select ename,sal from emp where sal BETWEEN 1000 and 2000;`. The result is a table with 2 columns: ename and sal. There are 6 rows of data displayed. Below the table, it says "6 rows in set (0.04 sec)".

ename	sal
ALLEN	1600.00
WARD	1250.00
MARTIN	1250.00
TURNER	1500.00
ADAMS	1100.00
MILLER	1300.00

Use of LIKE Special operator : (use of %)

Q21 : List the employees whose names start with “S” and (not “s”).

Ans : select ename from emp where ename LIKE "S%";

Q22 : List the employee names ending with an “S”.

Ans: select ename from emp where ename LIKE "%S";

```
MySQL 5.5 Command Line Client
mysql> select ename from emp where ename LIKE "S%";
+-----+
| ename |
+-----+
| SCOTT |
| SMITH |
+-----+
2 rows in set (0.00 sec)

mysql> select ename from emp where ename LIKE "%S";
+-----+
| ename |
+-----+
| ADAMS |
| JAMES |
| JONES |
+-----+
3 rows in set (0.01 sec)
```

Use of LIKE Special operator : (use of _)

Q23: List the names of employees whose names have exactly 5 characters.

Ans: select ename from emp where ename LIKE "_____";

Q24: List the names of employee names having “I” as the second character.

Ans: select ename from emp where ename LIKE "_I%";

```
MySQL 5.5 Command Line Client
mysql> select ename from emp where ename LIKE "_____";
+-----+
| ename |
+-----+
| ADAMS |
| ALLEN |
| BLAKE |
| CLARK |
| JAMES |
| JONES |
| SCOTT |
| SMITH |
+-----+
8 rows in set (0.01 sec)

mysql> select ename from emp where ename LIKE "_I%";
+-----+
| ename |
+-----+
| KING  |
| MILLER|
+-----+
2 rows in set (0.00 sec)
```

Working with NULL values:

Q25: List the names of employees who are not eligible for commission.

Ans: select ename from emp where comm is NULL;

```
MySQL 5.5 Command Line Client
mysql> select ename from emp where comm is NULL;
+-----+
| ename |
+-----+
| SMITH |
| JONES |
| BLAKE |
| CLARK |
| SCOTT |
| KING  |
| ADAMS |
| JAMES |
| FORD  |
| MILLER|
+-----+
10 rows in set (0.00 sec)

mysql> _
```

Q26: List the names of employees and designation of employee who does not report to anybody.

Ans: select ename, job from emp where mgr is NULL;

```
MySQL 5.5 Command Line Client
mysql> select ename, job from emp where mgr is NULL;
+-----+-----+
| ename | job      |
+-----+-----+
| KING  | PRESIDENT|
+-----+-----+
1 row in set (0.00 sec)

mysql>
```

Q27: List the employees who are eligible for commission.

Ans: select ename from emp where comm is NOT NULL;

```
MySQL 5.5 Command Line Client
mysql> select ename from emp where comm is NOT NULL;
+-----+
| ename |
+-----+
| ALLEN |
| WARD  |
| MARTIN|
| TURNER|
+-----+
4 rows in set (0.00 sec)
```

Q28: List the details of employees whose salary is greater than 2000 and commission is NULL.

Ans: select * from emp where sal > 2000 AND comm IS NULL;

```
MySQL 5.5 Command Line Client
mysql> select * from emp where sal > 2000 AND comm IS NULL;
```

eno	ename	job	mgr	hiredate	sal	comm	dno
7566	JONES	MANAGER	7839	1981-04-02	2975.00	NULL	20
7698	BLAKE	MANAGER	7839	1981-05-01	2850.00	NULL	30
7782	CLARK	MANAGER	7839	1981-06-09	2450.00	NULL	10
7788	SCOTT	ANALYST	7566	1987-04-19	3000.00	NULL	20
7839	KING	PRESIDENT	NULL	1981-11-17	5000.00	NULL	10
7902	FORD	ANALYST	7566	1981-12-03	3000.00	NULL	20

6 rows in set (0.00 sec)

Working with Alias values : (Derived attribute)

Q29: List the ename ,salary and PF amount of all employees.(PF is 10% of sal)

Ans: select ename, sal, sal*.1 from emp;

```
MySQL 5.5 Command Line Client
mysql> select ename, sal, sal*.1 from emp;
```

ename	sal	sal*.1
SMITH	800.00	80.00
ALLEN	1600.00	160.00
WARD	1250.00	125.00
JONES	2975.00	297.50
MARTIN	1250.00	125.00
BLAKE	2850.00	285.00
CLARK	2450.00	245.00
SCOTT	3000.00	300.00
KING	5000.00	500.00
TURNER	1500.00	150.00
ADAMS	1100.00	110.00
JAMES	950.00	95.00
FORD	3000.00	300.00
MILLER	1300.00	130.00

14 rows in set (0.02 sec)

Q30: write same above query using alias....

Ans: select ename, sal, sal*.1 PF from emp;

```
MySQL 5.5 Command Line Client
mysql> select ename, sal, sal*.1 PF from emp;
```

ename	sal	PF
SMITH	800.00	80.00
ALLEN	1600.00	160.00
WARD	1250.00	125.00
JONES	2975.00	297.50
MARTIN	1250.00	125.00
BLAKE	2850.00	285.00
CLARK	2450.00	245.00
SCOTT	3000.00	300.00
KING	5000.00	500.00
TURNER	1500.00	150.00
ADAMS	1100.00	110.00
JAMES	950.00	95.00
FORD	3000.00	300.00
MILLER	1300.00	130.00

14 rows in set (0.00 sec)

Working with ORDER BY clause :

Q31: List the eno,ename ,salary in ascending order of salary.

Ans: select eno,ename, sal from emp order by sal;

MySQL 5.5 Command Line Client

```
mysql> select eno,ename, sal from emp order by sal;
```

eno	ename	sal
7369	SMITH	800.00
7900	JAMES	950.00
7876	ADAMS	1100.00
7521	WARD	1250.00
7654	MARTIN	1250.00
7934	MILLER	1300.00
7844	TURNER	1500.00
7499	ALLEN	1600.00
7782	CLARK	2450.00
7698	BLAKE	2850.00
7566	JONES	2975.00
7902	FORD	3000.00
7788	SCOTT	3000.00
7839	KING	5000.00

14 rows in set (0.00 sec)

Q32: List the eno,ename ,salary in descending order of salary.

Ans: select eno,ename, sal from emp order by sal DESC;

MySQL 5.5 Command Line Client

```
mysql> select eno,ename, sal from emp order by sal DESC;
```

eno	ename	sal
7839	KING	5000.00
7788	SCOTT	3000.00
7902	FORD	3000.00
7566	JONES	2975.00
7698	BLAKE	2850.00
7782	CLARK	2450.00
7499	ALLEN	1600.00
7844	TURNER	1500.00
7934	MILLER	1300.00
7654	MARTIN	1250.00
7521	WARD	1250.00
7876	ADAMS	1100.00
7900	JAMES	950.00
7369	SMITH	800.00

14 rows in set (0.00 sec)

Q33: List the dno,job,ename ,salary in ascending order of dno and then descending order of salary.

Ans: select dno,job,ename, sal from emp order by dno, sal DESC;

```

MySQL 5.5 Command Line Client

mysql> select dno,job,ename, sal from emp order by dno, sal DESC;
+-----+-----+-----+-----+
| dno | job | ename | sal |
+-----+-----+-----+-----+
| 10 | PRESIDENT | KING | 5000.00 |
| 10 | MANAGER | CLARK | 2450.00 |
| 10 | CLERK | MILLER | 1300.00 |
| 20 | ANALYST | FORD | 3000.00 |
| 20 | ANALYST | SCOTT | 3000.00 |
| 20 | MANAGER | JONES | 2975.00 |
| 20 | CLERK | ADAMS | 1100.00 |
| 20 | CLERK | SMITH | 800.00 |
| 30 | MANAGER | BLAKE | 2850.00 |
| 30 | SALESMAN | ALLEN | 1600.00 |
| 30 | SALESMAN | TURNER | 1500.00 |
| 30 | SALESMAN | MARTIN | 1250.00 |
| 30 | SALESMAN | WARD | 1250.00 |
| 30 | CLERK | JAMES | 950.00 |
+-----+-----+-----+-----+
14 rows in set (0.00 sec)

```

Q34: List the eno, ename ,salary and PF (10% of sal) in ascending order of PF.

Ans: select eno,ename, sal ,sal * .1 PF from emp order by 4;

```

MySQL 5.5 Command Line Client

mysql> select eno,ename,sal,sal*.1 PF from emp order by 4;
+-----+-----+-----+-----+
| eno | ename | sal | PF |
+-----+-----+-----+-----+
| 7369 | SMITH | 800.00 | 80.00 |
| 7900 | JAMES | 950.00 | 95.00 |
| 7876 | ADAMS | 1100.00 | 110.00 |
| 7521 | WARD | 1250.00 | 125.00 |
| 7654 | MARTIN | 1250.00 | 125.00 |
| 7934 | MILLER | 1300.00 | 130.00 |
| 7844 | TURNER | 1500.00 | 150.00 |
| 7499 | ALLEN | 1600.00 | 160.00 |
| 7782 | CLARK | 2450.00 | 245.00 |
| 7698 | BLAKE | 2850.00 | 285.00 |
| 7566 | JONES | 2975.00 | 297.50 |
| 7902 | FORD | 3000.00 | 300.00 |
| 7788 | SCOTT | 3000.00 | 300.00 |
| 7839 | KING | 5000.00 | 500.00 |
+-----+-----+-----+-----+
14 rows in set (0.00 sec)

```

Q35: List the eno, ename ,salary and PF (10% of sal) in ascending order of PF.

Ans: select eno,ename, sal ,sal * .1 PF from emp order by Sal*.1;

```

MySQL 5.5 Command Line Client

mysql> select eno,ename,sal,sal*.1 PF from emp order by sal*.1;
+-----+-----+-----+-----+
| eno | ename | sal | PF |
+-----+-----+-----+-----+
| 7369 | SMITH | 800.00 | 80.00 |
| 7900 | JAMES | 950.00 | 95.00 |
| 7876 | ADAMS | 1100.00 | 110.00 |
| 7521 | WARD | 1250.00 | 125.00 |
| 7654 | MARTIN | 1250.00 | 125.00 |
| 7934 | MILLER | 1300.00 | 130.00 |
| 7844 | TURNER | 1500.00 | 150.00 |
| 7499 | ALLEN | 1600.00 | 160.00 |
| 7782 | CLARK | 2450.00 | 245.00 |
| 7698 | BLAKE | 2850.00 | 285.00 |
| 7566 | JONES | 2975.00 | 297.50 |
| 7902 | FORD | 3000.00 | 300.00 |
| 7788 | SCOTT | 3000.00 | 300.00 |
| 7839 | KING | 5000.00 | 500.00 |
+-----+-----+-----+-----+
14 rows in set (0.02 sec)

```

Q36 : Or

Ans: select eno,ename, sal ,sal * .1 PF from emp order by PF;

```

MySQL 5.5 Command Line Client

mysql> select eno,ename,sal,sal*.1 PF from emp order by PF;
+-----+-----+-----+-----+
| eno   | ename  | sal    | PF    |
+-----+-----+-----+-----+
| 7369  | SMITH  | 800.00 | 80.00 |
| 7900  | JAMES  | 950.00 | 95.00 |
| 7876  | ADAMS  | 1100.00 | 110.00 |
| 7521  | WARD   | 1250.00 | 125.00 |
| 7654  | MARTIN | 1250.00 | 125.00 |
| 7934  | MILLER | 1300.00 | 130.00 |
| 7844  | TURNER | 1500.00 | 150.00 |
| 7499  | ALLEN  | 1600.00 | 160.00 |
| 7782  | CLARK  | 2450.00 | 245.00 |
| 7698  | BLAKE  | 2850.00 | 285.00 |
| 7566  | JONES  | 2975.00 | 297.50 |
| 7902  | FORD   | 3000.00 | 300.00 |
| 7788  | SCOTT  | 3000.00 | 300.00 |
| 7839  | KING   | 5000.00 | 500.00 |
+-----+-----+-----+-----+
14 rows in set (0.00 sec)

```

Conclusion:

R1	R2	R3	R4	Total	Sign with Date
(3)	(5)	(4)	(3)	(15)	

DML (Data Manipulation Language) Queries on Case study

A. Student Table (INSERT, UPDATE, DELETE, SELECT)

1. Add a new student to the system.

```
11 |
12 | INSERT INTO Student (Student_ID, Name, Email, Phone, CGPA, Skills, Resume)
13 | VALUES (4, 'Neha Kapoor', 'neha@gmail.com', '9876543213', 8.2, 'Python, Machine Learning', 'neha_resume.pdf');
14 |
```

Student_ID	Name	Email	Phone	CGPA	Skills	Resume
4	Neha Kapoor	neha@gmail.com	9876543213	8.20	Python, Machine Learning	neha_resume.pdf

2. Update the CGPA of a student.

```
15 | UPDATE Student
16 | SET CGPA = 8.9
17 | WHERE Student_ID = 2;
18 |
```

Student_ID	Name	Email	Phone	CGPA	Skills	Resume
2	Priya Verma	priya@gmail.com	9876543211	8.90	HTML, CSS, JavaScript	priya_resume.pdf

3. Delete a student who is no longer participating in placements.

```
19 |
20 | DELETE FROM Student
21 | WHERE Student_ID = 3;
22 |
```

Student_ID	Name	Email	Phone	CGPA	Skills	Resume
1	Rahul Sharma	rahul@gmail.com	9876543210	8.50	Java, SQL, Python	rahul_resume.pdf
2	Priya Verma	priya@gmail.com	9876543211	9.00	HTML, CSS, JavaScript	priya_resume.pdf
4	Sneha Joshi	sneha@gmail.com	9876543213	8.20	Python, Data Science, ML	sneha_resume.pdf
5	Rohan Desai	rohan@gmail.com	9876543214	7.50	JavaScript, React, Node.js	rohan_resume.pdf
6	Anjali Mehta	anjali@gmail.com	9876543215	8.90	Java, Spring Boot, Microservices	anjali_resume.pdf
7	Vikram Singh	vikram@gmail.com	9876543216	8.00	DevOps, AWS, Docker	vikram_resume.pdf

4. Retrieve all students who have a CGPA above 8.0.

```
21 |
22 | SELECT * FROM Student
23 | WHERE CGPA > 8.0;
24 |
```

Student_ID	Name	Email	Phone	CGPA	Skills	Resume
1	Rahul Sharma	rahul@gmail.com	9876543210	8.50	Java, SQL, Python	rahul_resume.pdf
2	Priya Verma	priya@gmail.com	9876543211	9.00	HTML, CSS, JavaScript	priya_resume.pdf
4	Sneha Joshi	sneha@gmail.com	9876543213	8.20	Python, Data Science, ML	sneha_resume.pdf
6	Anjali Mehta	anjali@gmail.com	9876543215	8.90	Java, Spring Boot, Microservices	anjali_resume.pdf

B. Company Table (INSERT, UPDATE, DELETE, SELECT)

1. Insert a new company into the system.

```
11
12 INSERT INTO Company (Company_ID, Name, Industry, Contact_Person, Email)
13 VALUES (106, 'Wipro', 'IT', 'Akash Gupta', 'wipro@company.com');
14
```

Company_ID	Name	Industry	Contact_Person	Email
106	Wipro	IT	Akash Gupta	wipro@company.com

2. Change the contact person for Google.

```
14
15 UPDATE Company
16 SET Contact_Person = 'Alice Johnson'
17 WHERE Company_ID = 101;
18
```

Company_ID	Name	Industry	Contact_Person	Email
101	Google	IT	Alice Johnson	google@company.com

3. Remove a company from the system.

```
17
18 DELETE FROM Company
19 WHERE Company_ID = 103;
20
```

Company_ID	Name	Industry	Contact_Person	Email
101	Google	IT	John Doe	google@company.com
102	Microsoft	Software	Jane Smith	microsoft@company.com
104	Amazon	E-commerce	Samantha Wilson	amazon@company.com
105	Infosys	Technology	Anil Mehta	infosys@company.com

4. Retrieve all companies in the IT industry.

```
15  
20 SELECT * FROM Company  
21 WHERE Industry = 'IT';  
22
```

Company_ID	Name	Industry	Contact_Person	Email
101	Google	IT	John Doe	google@company.com
106	Wipro	IT	Akash Gupta	wipro@company.com

C. Job Table (INSERT, UPDATE, DELETE, SELECT)

1. Insert a new job posting for a Data Scientist role.

```
15  
16 INSERT INTO Job (Job_ID, Job_Title, Description, Eligibility, Salary, Company_ID)  
17 VALUES (207, 'Data Scientist', 'Analyze and interpret complex data', 'CGPA >= 8.0, Python, SQL, ML', 1500000, 104);  
18
```

Job_ID	Job_Title	Description	Eligibility	Salary	Company_ID
207	Data Scientist	Analyze and interpret complex data	CGPA >= 8.0, Python, SQL, ML	1500000.00	104

2. Increase the salary of all Software Engineer roles by 10%.

```
19 UPDATE Job  
20 SET Salary = Salary * 1.10  
21 WHERE Job_Title = 'Software Engineer';  
22
```

Job_ID	Job_Title	Description	Eligibility	Salary	Company_ID
201	Software Engineer	Develop web applications	CGPA >= 8.0, Java, SQL	1320000.00	101

3. Remove a job posting that is no longer available.

```
15  
16 DELETE FROM Job  
17 WHERE Job_ID = 204;  
18  
19 SELECT * FROM Job;  
20
```

Job_ID	Job_Title	Description	Eligibility	Salary	Company_ID
201	Software Engineer	Develop web applications	CGPA >= 8.0, Java, SQL	1200000.00	101
202	Data Analyst	Analyze company data	CGPA >= 7.5, Python, SQL	1000000.00	102
203	Cloud Engineer	Manage cloud services	CGPA >= 8.0, AWS, Linux	1100000.00	103
205	Machine Learning Engineer	Build AI models and systems	CGPA >= 8.5, Python, ML, TensorFlow	1300000.00	101
206	Frontend Developer	Develop user-friendly interfaces	CGPA >= 7.0, HTML, CSS, JavaScript, React	950000.00	105

4. Retrieve all job postings offering a salary above 1,000,000.

```
23  
24 SELECT * FROM Job  
25 WHERE Salary > 1000000;  
26
```

Job_ID	Job_Title	Description	Eligibility	Salary	Company_ID
201	Software Engineer	Develop web applications	CGPA >= 8.0, Java, SQL	1200000.00	101
203	Cloud Engineer	Manage cloud services	CGPA >= 8.0, AWS, Linux	1100000.00	103
204	Cybersecurity Analyst	Monitor and secure networks	CGPA >= 7.8, Cybersecurity, Networking	1050000.00	104
205	Machine Learning Engineer	Build AI models and systems	CGPA >= 8.5, Python, ML, TensorFlow	1300000.00	101

D. Interview Table (INSERT, UPDATE, DELETE, SELECT)

1 Insert a new interview schedule.

```
20  
21 INSERT INTO Interview (Interview_ID, Student_ID, Job_ID, Interview_Date, Result)  
22 VALUES (4, 1, 204, '2025-04-20', 'Pending');  
23
```

Interview_ID	Student_ID	Job_ID	Interview_Date	Result
4	1	204	2025-04-20	Pending

2. Update the interview result of a student.

```
16  
17 UPDATE Interview  
18 SET Result = 'Passed'  
19 WHERE Interview_ID = 2;  
20
```

Interview_ID	Student_ID	Job_ID	Interview_Date	Result
2	2	202	2025-04-12	Passed

3. Remove an interview record.

```
19  
20 DELETE FROM Interview  
21 WHERE Interview_ID = 3;  
22
```

Interview_ID	Student_ID	Job_ID	Interview_Date	Result
1	1	201	2025-04-10	Pending
2	2	202	2025-04-12	Passed
4	4	204	2025-04-18	Passed
5	5	205	2025-04-20	Pending
6	6	206	2025-04-22	Failed
7	7	201	2025-04-25	Passed

4. Retrieve all scheduled interviews for a specific date.

```
20  
21 SELECT * FROM Interview  
22 WHERE Interview_Date = '2025-04-10';  
23
```

Interview_ID	Student_ID	Job_ID	Interview_Date	Result
1	1	201	2025-04-10	Pending

E. Placement Officer Table (INSERT, UPDATE, DELETE, SELECT)

1. Insert a new placement officer.

```
10  
11 INSERT INTO Placement_Officer (Officer_ID, Name, Email, Phone)  
12 VALUES (504, 'Suman Joshi', 'suman@college.com', '9876543222');  
13
```

Officer_ID	Name	Email	Phone
504	Suman Joshi	suman@college.com	9876543222

2. Update the contact information of a placement officer.

```
15 UPDATE Placement_Officer
16 SET Phone = '9876543333'
17 WHERE Officer_ID = 502;
```

Officer_ID	Name	Email	Phone
502	Anita Mehta	anita@college.com	9876543333

3. Remove a placement officer record.

```
10
11
12 DELETE FROM Placement_Officer
13 WHERE Officer_ID = 501;
```

Officer_ID	Name	Email	Phone
502	Anita Mehta	anita@college.com	9876543221
503	Sunil Desai	sunil@college.com	9876543222

4. Retrieve all placement officers handling interviews.

```
18 |
19 SELECT * FROM Placement_Officer
20 WHERE Name = 'Anita Mehta';
21
```

Officer_ID	Name	Email	Phone
502	Anita Mehta	anita@college.com	9876543221